

Compositional monitoring of energy-mix drift on the simplex

Which carrier did the structural work?

Standard compositional data analysis — what happens when we read the simplex (shares that sum to one) forward in time.

Follow along on the repository — the slide deck, manuscript, and live projector are all open.

H^s runs any compositional dataset the CoDa community can describe — every view here is reproducible on your data.

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github.com / PeterHiggins19 / higgins-decomposition

Community folder: CODA-Association/ · Hand-out in UN-6 locales (EN · FR · ES · RU · ZH · AR)

The instrument reads. The expert decides. The hashes carry the receipts. The vocabulary holds the line.

The size view hides the work

A carrier can be small in share yet do most of the structural work — its share of the squared compositional motion.

**Germany electricity, 25 years
the size view — the standard share / stacked-area chart**

Coal and lignite recede. Nuclear phases out.
Gas holds. Solar and wind grow steadily.
Wind passes coal in the late 2010s.

What the size view misses

Germany Solar, 2005 → 2006

starting share	0.21 %
structural Power Share	71.1 %
Activation Coefficient	$\approx 333 \times$

*Solar acted at $333 \times$ its size,
four years before the share view
calls solar visible.*

Deceptive drift: the size view hides which carrier did the structural work — here a 0.21 % carrier does 71 % of it.

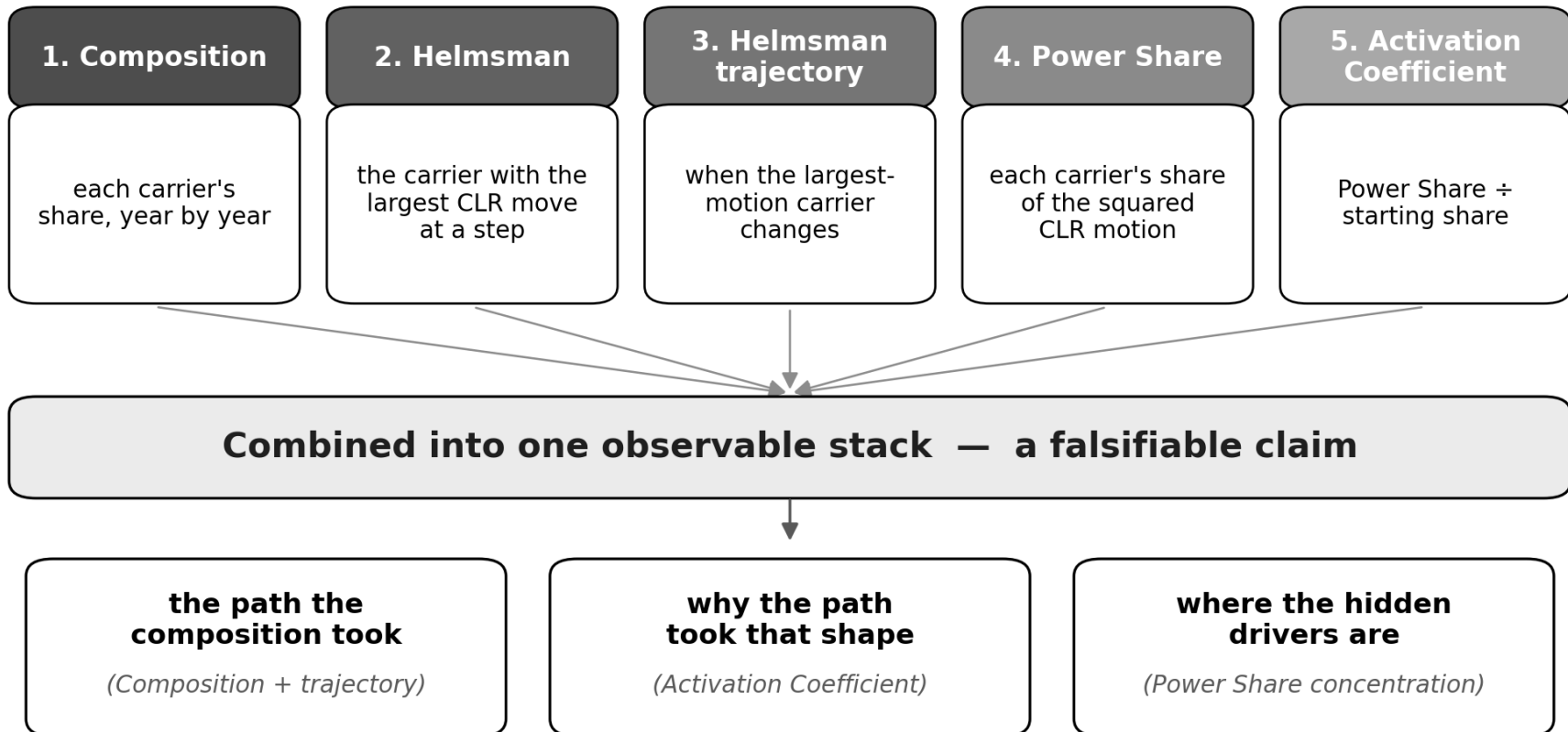
Mathematics: standard compositional data analysis. Application: monitoring frame may be new.

Reading time on the simplex

Standard CoDa geometry, read forward in time — five named readings, each defined below.

INPUT · EMBER electricity-generation data · carriers × N years per country

Closure → Centred log-ratio (CLR) → Helmert-orthonormal ILR → Aitchison-distance trajectory



Five readings · one observable stack · reproducible from the raw CSV

The Activation Coefficient

How much structural work a carrier does, relative to how much of the mix it is.

$$\alpha_i(t) = \frac{\text{Power Share}_i(t)}{\text{starting share}_i(t)}$$

$\alpha \approx 1$ carrier does work proportional to its size — ordinary

$\alpha \gg 1$ carrier acts far above its size — hidden driver

$\alpha < 1$ carrier carries less work than its size suggests — coasting

Worked example — Germany Solar 2005 → 2006

starting share	0.21 %	small
Power Share	71.1 %	most of the work
α	$\approx 333 \times$	hidden driver

Analogy — yeast is 2 % of a loaf by mass and does 100 % of the rising: the same shape.

The Energiewende's structural beginning — four years before solar appears in the share view.

Three archetypes — one instrument, three regimes

The same protocol on three transitions that look fundamentally different.

Germany

deliberate transition

continuous arc

Energiewende
2000 → 2025
solar + wind absorb
structural work
before size dominates.

Japan

external shock

loop and reorganise

Fukushima 2011
displaces nuclear,
causes 2011–2013
multi-year
reorganisation.

United Kingdom

regime change

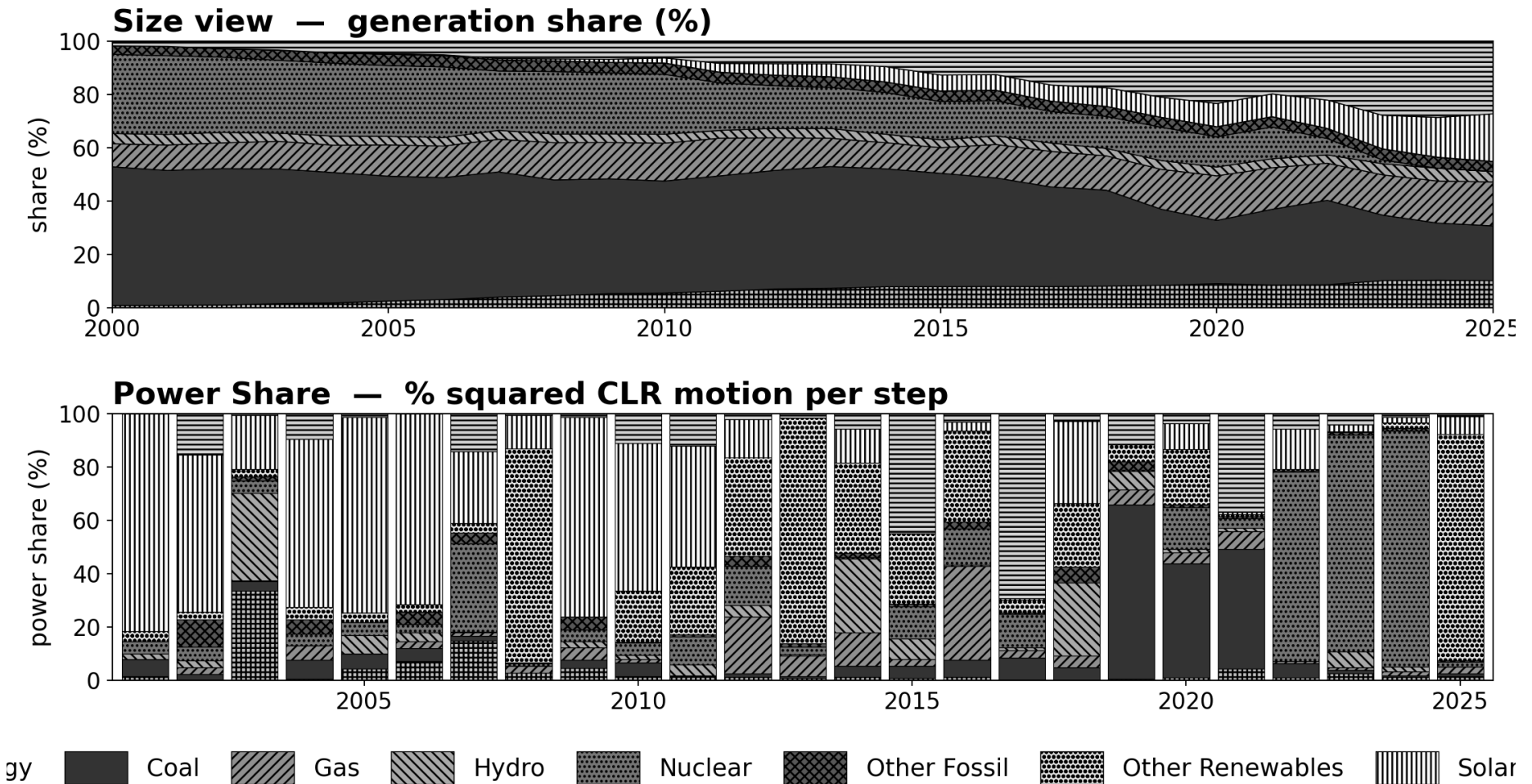
jump and return

Coal exit 2012–2020
from > 30 % to < 2 %.
Wind, solar, others
absorb displaced
structural work.

Three different transition regimes. One operational protocol reads them all.

Germany — share and structural work

The Energiewende read as a single smooth arc on the simplex.



Solar 2005–2006: 0.21 % share • 71.1 % structural work • $\alpha \approx 333 \times$

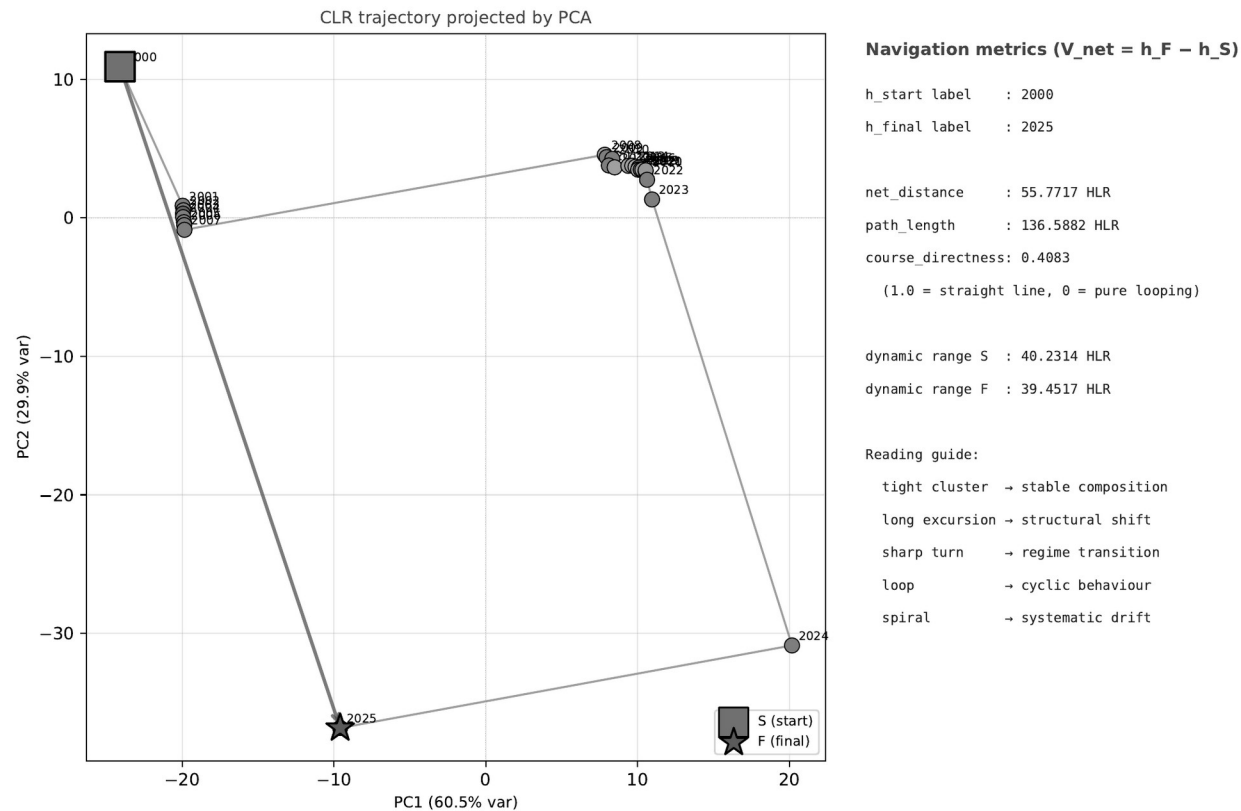
Tiny share, dominant work — the deceptive-drift signature, four years before the share view shows anything.

Germany — the trajectory

The Helmsman trajectory: where the composition went, year by year.

Compositional system geometry and dynamics for, Germany — EMBER electricity generation 2000-2025

Plate 16 | System Course Plot — PCA 2D path with S→F net vector
Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

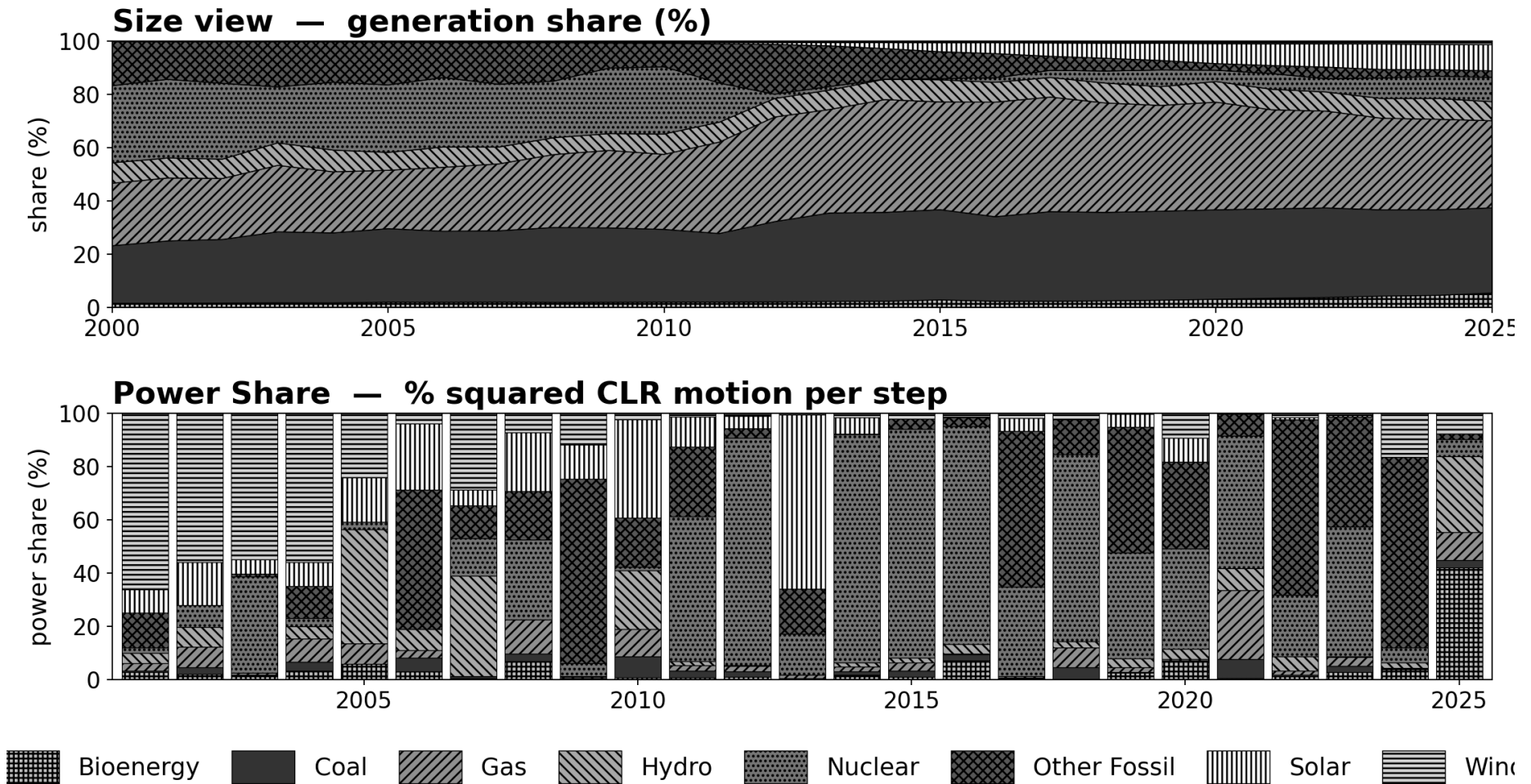


Trajectory directness 0.41 (end-to-end distance ÷ path length) — a continuous arc toward the renewable vertex.

Smooth, monotone reorientation — no loops, no flips. Deliberate transition as a sustained trajectory.

Japan — share and structural work

Fukushima 2011: an external shock displaces nuclear; a multi-year reorganisation follows.



Aitchison distance (distance on the simplex) 2011 → 2012 $\approx 3 \times$ the baseline step • 17 Helmsman flips

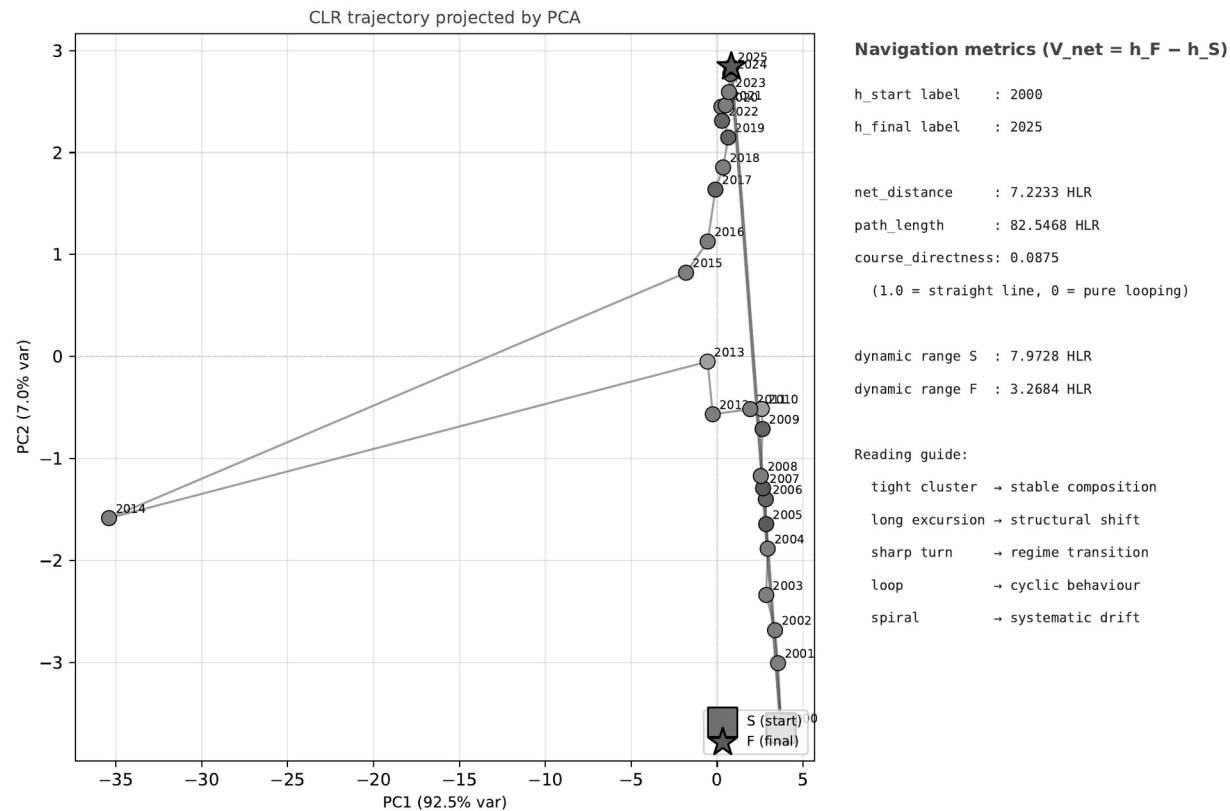
The instrument detects both the shock and the multi-year reorganisation that followed.

Japan — the trajectory

The Helmsman trajectory: looping reorganisation, not a single step.

Compositional system geometry and dynamics for, Japan — EMBER electricity generation 2000-2025

Plate 16 | System Course Plot — PCA 2D path with S→F net vector
Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



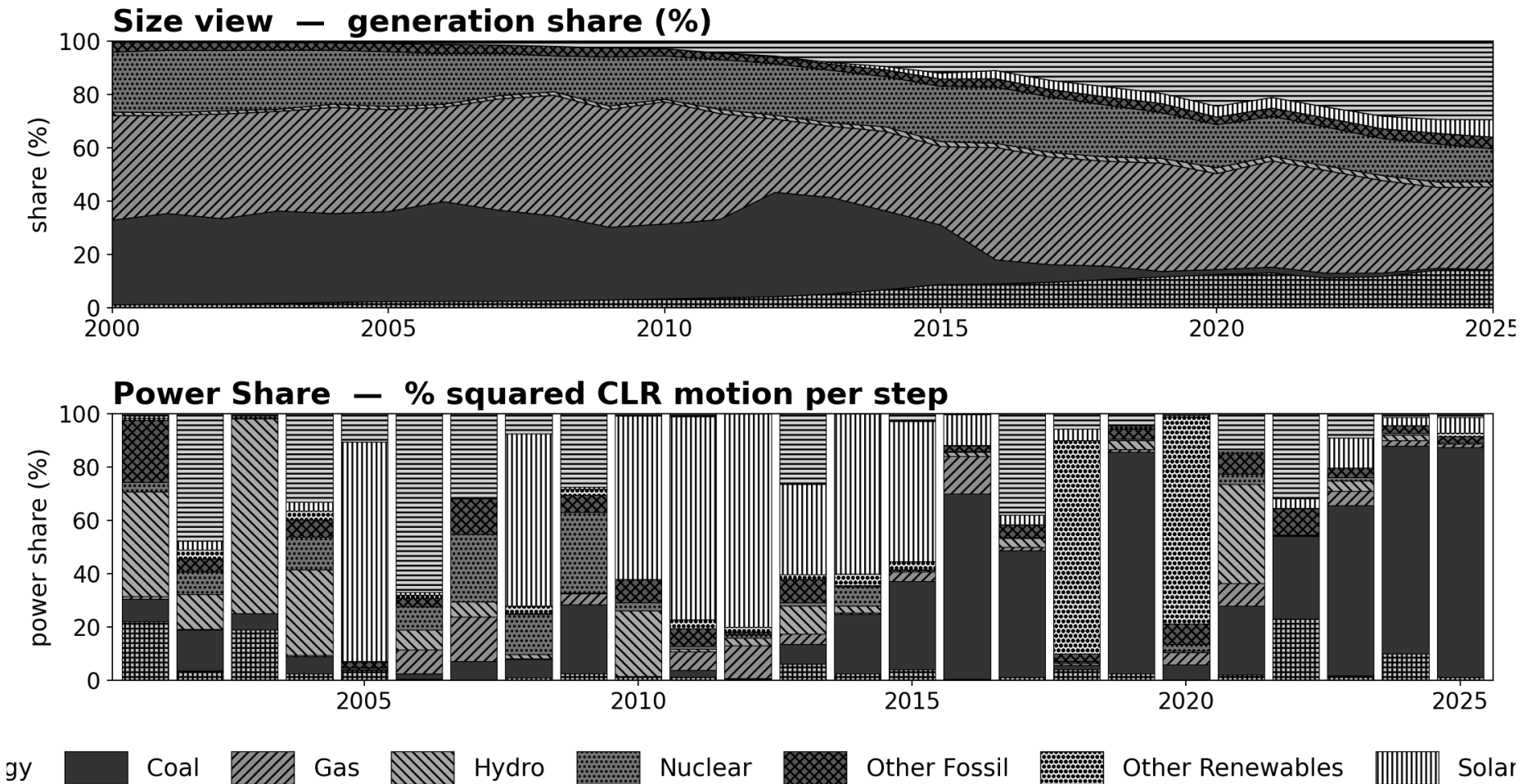
engine: cnt 2.0.4 (c5111ef761bb...) | data: f61dcf0026fb... | source: e7643dca648e...
2026-05-06 | run: v1.1.x-final | page 16/28 | STAGE 2 (Order 2) | system course plot (PCA)

Trajectory directness 0.09 — the loop-and-reorganise pattern.

The trajectory revisits and reroutes — the system searches for a new composition, not a planned trajectory.

United Kingdom — share and structural work

Coal exit as policy-driven regime change, absorbed across several carriers.



Coal: > 30 % → < 2 %. Wind, solar, and other renewables absorb the displaced work.

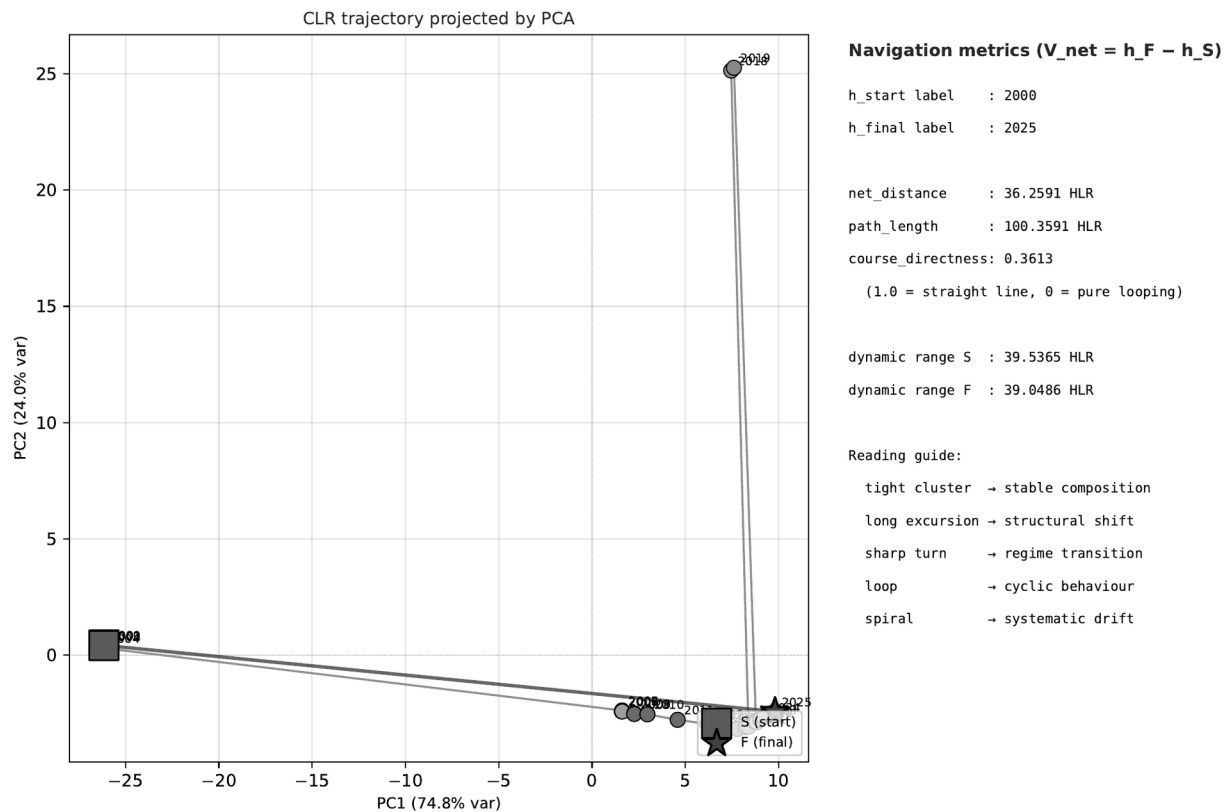
Displaced work redistributes across several carriers — not one substitute, but several.

United Kingdom — the trajectory

The Helmsman trajectory: a jump from the coal vertex, then a return toward stability.

Compositional system geometry and dynamics for, United Kingdom — EMBER electricity generation 2000-2025

Plate 16 | System Course Plot — PCA 2D path with S→F net vector
Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



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2026-05-06 | run: v1.1.x-final | page 16/28 | STAGE 2 (Order 2) | system course plot (PCA)

Trajectory directness 0.36 — the jump-and-return pattern.

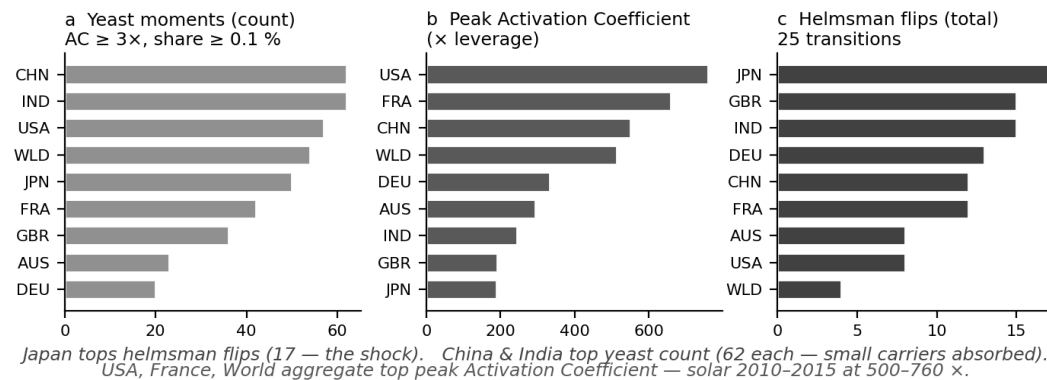
A sharp departure from the coal vertex, then re-stabilisation — regime change as one decisive displacement.

Across the corpus — deceptive drift in 5 of 9

The same protocol applied to all nine EMBER countries.

Deceptive drift: the size (share) view hides which carrier did the structural work — a small carrier carries a large share of the squared compositional motion (Activation Coefficient $\gg 1$).

Detected in five of nine countries; absent in four — a detector that reads real structure must not flag every system.



d Top ten activation moments across the 9-country corpus

#	country	transition	carrier	Activation	Power Share	Size at start
1.	USA	2012-2013	Solar	760 $\times$	81.7 %	0.107 %
2.	FRA	2010-2011	Solar	659 $\times$	72.6 %	0.110 %
3.	FRA	2004-2005	Wind	634 $\times$	66.0 %	0.104 %
4.	CHN	2013-2014	Solar	549 $\times$	84.7 %	0.154 %
5.	WLD	2010-2011	Solar	513 $\times$	77.7 %	0.151 %
6.	FRA	2005-2006	Wind	495 $\times$	83.5 %	0.169 %
7.	USA	2013-2014	Solar	395 $\times$	87.9 %	0.223 %
8.	CHN	2004-2005	Bioenergy	346 $\times$	39.7 %	0.115 %
9.	CHN	2003-2004	Bioenergy	342 $\times$	45.5 %	0.133 %
10.	FRA	2024-2025	Coal	342 $\times$	62.1 %	0.182 %

deceptive drift present: AUS • CHN • GBR • IND • JPN

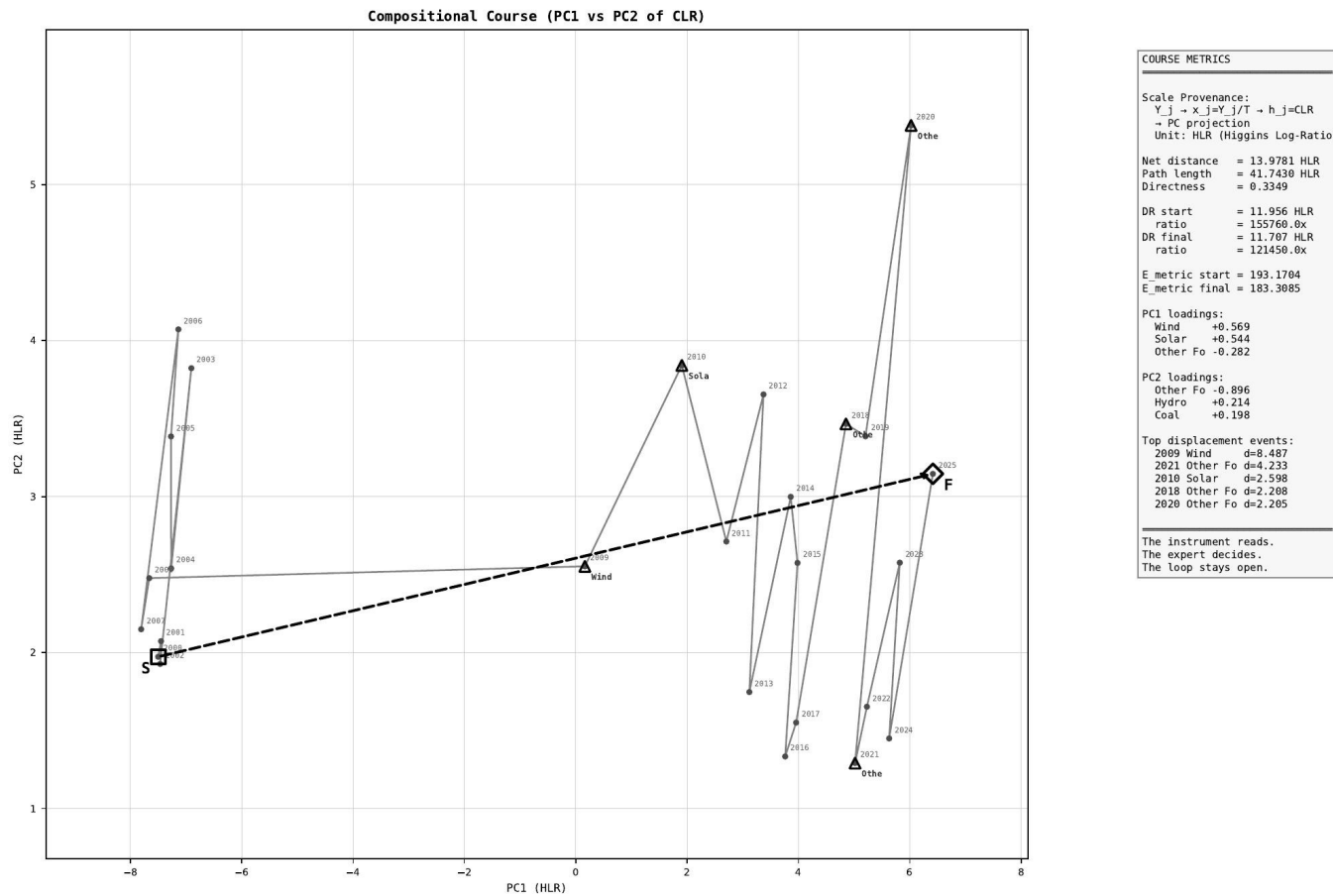
deceptive drift absent: DEU (annual) • FRA • USA • WLD

Discrimination — flagging some systems and not others — is itself evidence the detector reads real structure.

Australia — the trajectory

The rest of the world • deceptive drift present. Australia: the full 2000–2025 trajectory.

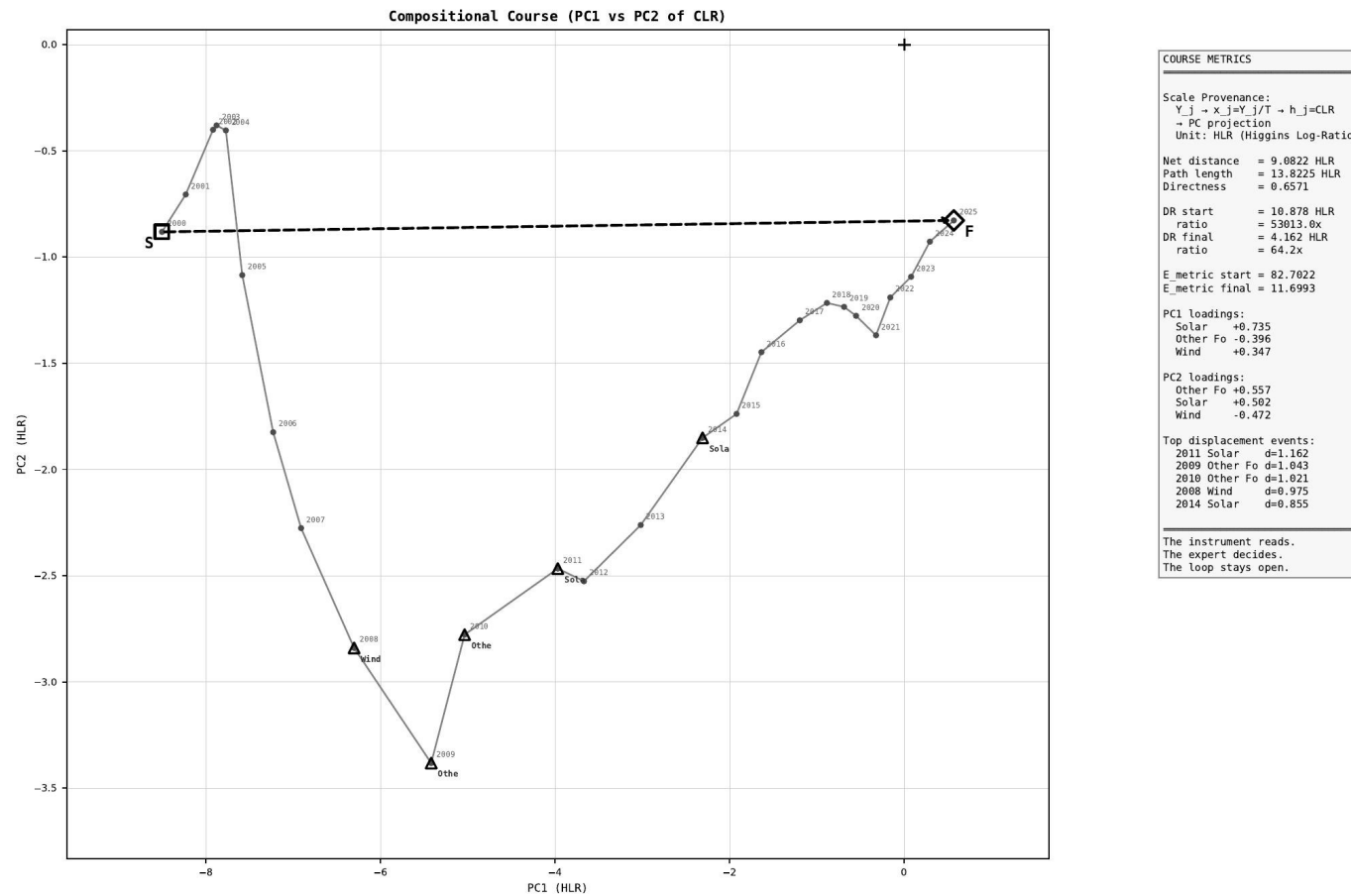
SYSTEM COURSE PLOT | ember_AUS_Australia_generation_TWh.csv | D=9 N=26 | Years 2000–2025



China — the trajectory

China: the full trajectory — coal-era growth, then a turn.

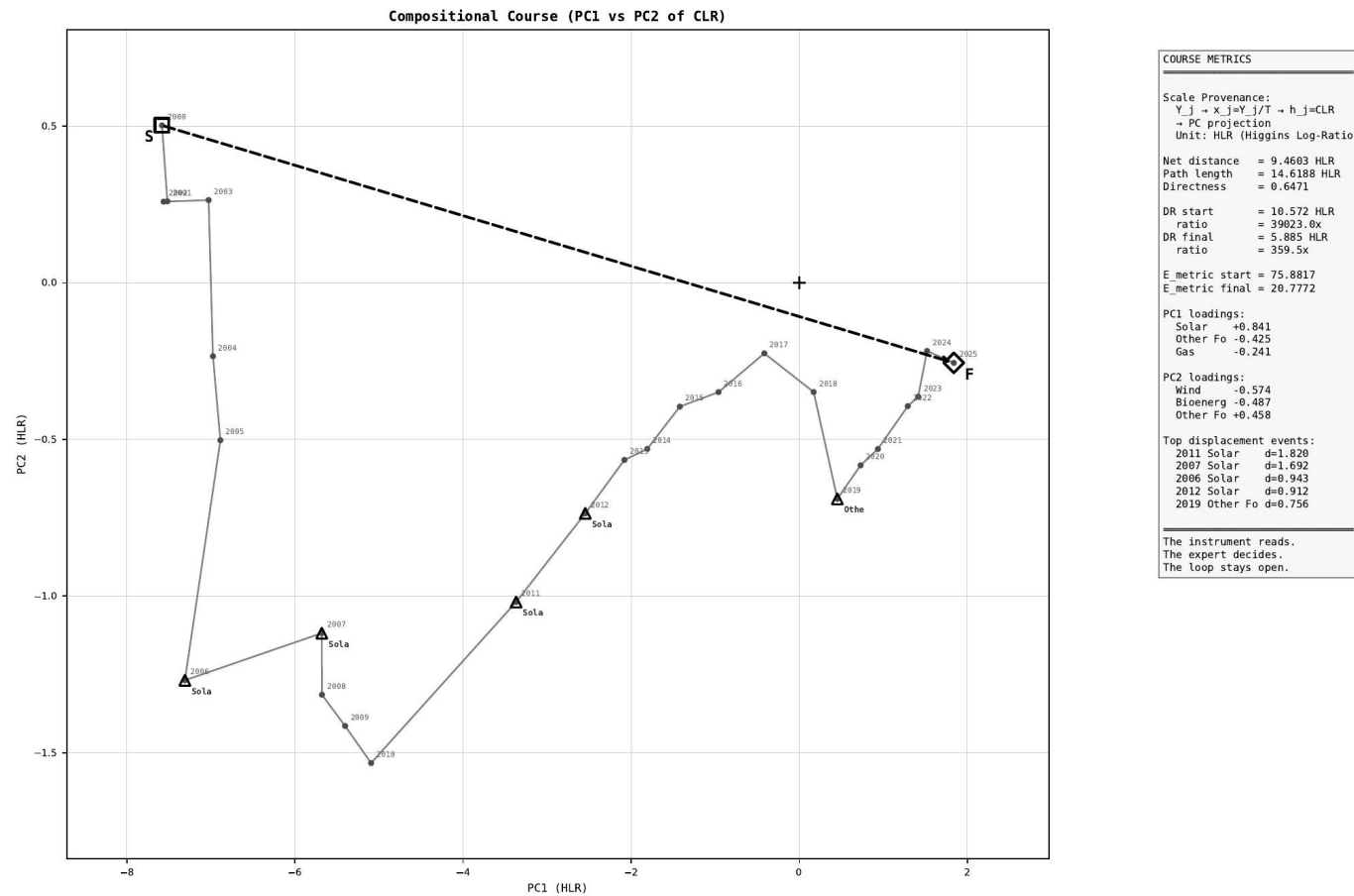
SYSTEM COURSE PLOT | ember_CHN_China_generation_TWh.csv | D=8 N=26 | Years 2000–2025



India — the trajectory

India: the full trajectory — the third case where deceptive drift is present.

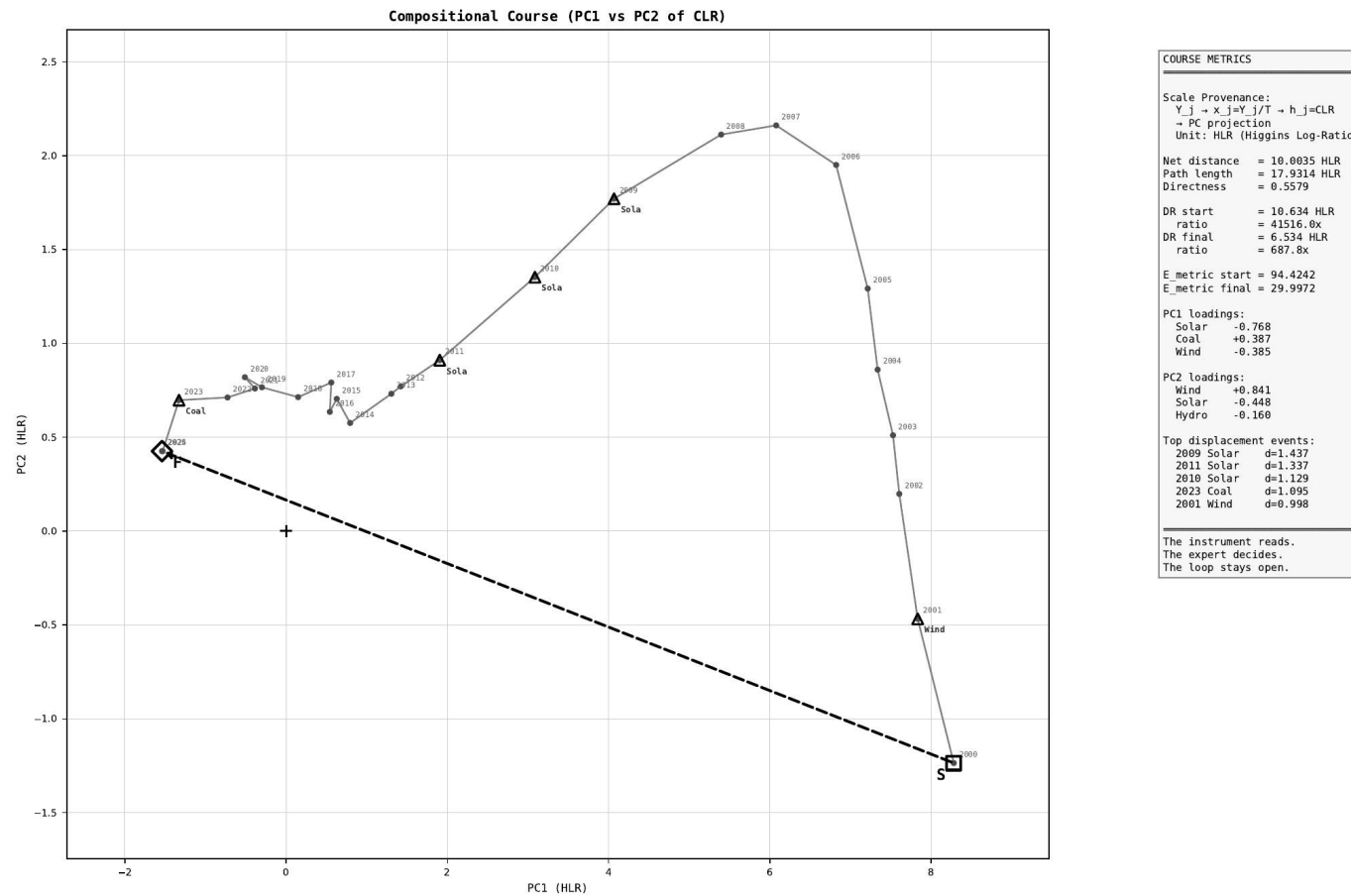
SYSTEM COURSE PLOT | ember_IND_India_generation_TWh.csv | D=8 N=26 | Years 2000–2025



France — the trajectory

Deceptive drift absent. France: the full trajectory — nuclear-stable.

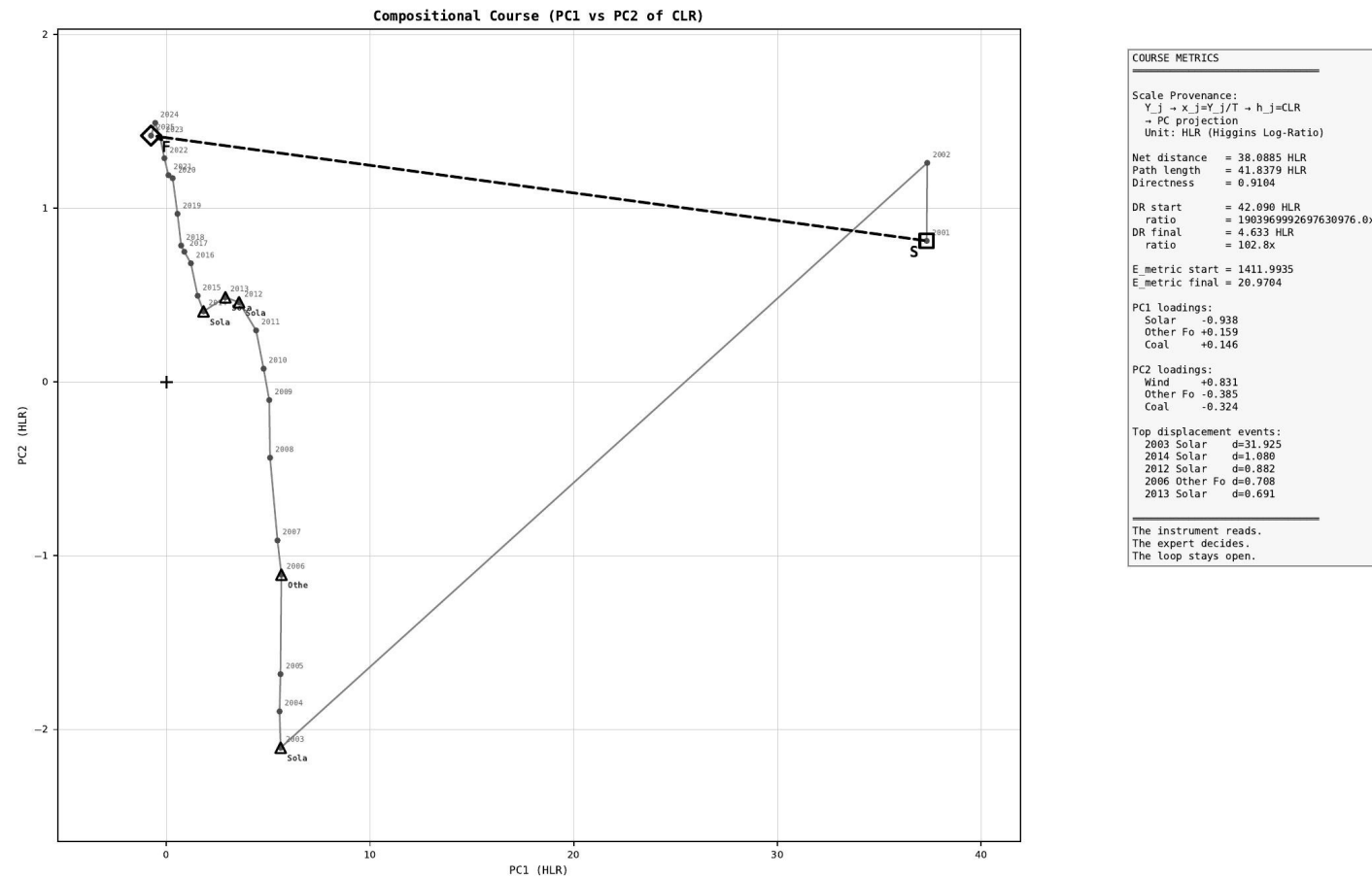
SYSTEM COURSE PLOT | ember_FRA_France_generation_TWh.csv | D=9 N=26 | Years 2000–2025



United States — the trajectory

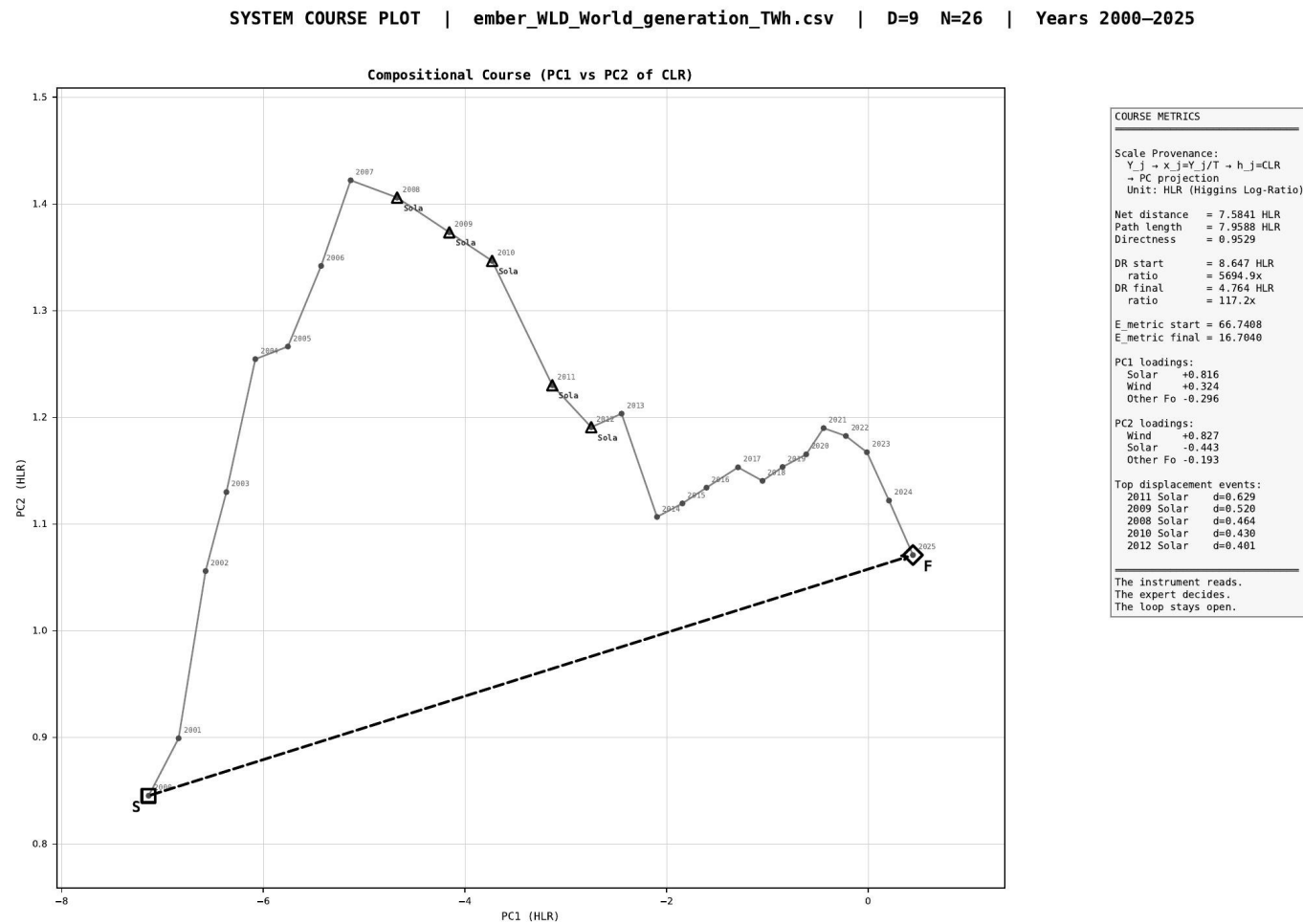
United States: the full trajectory — a large grid; no deceptive drift at annual grain.

SYSTEM COURSE PLOT | ember_USA_United_States_generation_TWh.csv | D=9 N=25 | Years 2001–2025



World — the trajectory

World: the aggregate trajectory — large-N smoothing hides the deceptive drift in its constituents.



What the stack answers

Recap of the five readings — then the live instrument takes over.

WHAT	carriers are big.	<i>size view</i>
WHO	is in largest motion.	<i>Helmsman</i>
WHEN	the largest-motion carrier changes.	<i>Helmsman trajectory</i>
HOW MUCH	structural work each carrier did.	<i>Power Share</i>
WHY	a small carrier mattered.	<i>Activation Coefficient</i>

The stack does not replace interpretation. It gives interpretation a reproducible object.

The live instrument takes it from here → [codawork2026_projector.html](#)

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